ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 12 m/s	v = at.
2	**(c)** 5 m/s	100/20.
3	**(d)** 30 m/s	t up = 3 s; u = gt.
4	**(b)** 100 m	s = (1/2)at squared.
5	**(d)** 48 km/h	Harmonic mean 2 x 40 x 60/100.
6	**(b)** 3.14 m/s	2 pi r/T = 12.56/4.
7	**(a)** 40 m	v squared = 2as.
8	**(d)** uniform acceleration from rest	v proportional to t.
9	**(d)** 2r	Diameter.
10	**(a)** 2.5 m/s squared	72 km/h = 20 m/s; a = 20/8.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	s = 15 x 20 = 300 m (1); Horizontal line; area gives distance (1).	1+1
12	t = 2 s (1); v = 20 m/s (1).	1+1
13	No air resistance (1); Acceleration due to gravity is the same for all (1).	1+1
14	a = 2 m/s squared (1); s = 9 x 3 = 27 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	a = 2 m/s squared (1); s = 25 x 5 = 125 m (1); Average velocity 25 m/s (1).	1+1+1
16	s_A = t squared; s_B = 20t (1); Equal when t = 20 s (1); Distance 400 m (1).	1+1+1
17	40 + 0 + 40 = 80 m (1); Time 20 s (1); Average speed 4 m/s (1).	1+1+1
	OR: Time = 2.5 s (1); H = 625/20 = 31.25 m (1); v at 1 s = 15 m/s (1).	1+1+1
18	a = 3 m/s squared (1); Area of triangle = 0.5 x 6 x 18 (1); = 54 m (1).	1+1+1
	OR: Velocity changes uniformly (1); Average v = (u + v)/2 (1); s = average velocity x t (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Maximum velocity 20 m/s (1); Retardation 4 m/s squared (1); Distances 100, 400 and 50 m (1); Total 550 m (1); Trapezium graph (1).	1+1+1+1+1
	OR: Speed 20 m/s (1); Reaction distance 10 m (1); Braking distance 40 m (1); Total 50 m (1); Just stops at the obstacle (1).	1+1+1+1+1
20	(a) 3 s (1); (b) 45 m (1); (c) 6 s (1); (d) 30 m/s downward (1); Symmetry (1).	1+1+1+1+1
	OR: Circumference = 2 x 3.14 x 3.84 x 10^5 = 2.41 x 10^6 km (1); Period = 27 x 86400 = 2.33 x 10^6 s (1); Speed = 1.03 km/s (1); Direction changes: acceleration (1); Speed constant (1).	1+1+1+1+1

Mistakes to avoid:

- Using distance instead of displacement (or speed instead of velocity); for a round trip the displacement and average velocity are zero.
- Not converting km/h to m/s before using the equations of motion, or mixing km with seconds.
- Putting a positive sign for retardation in $v = u + at$ or $v^2 = u^2 + 2as$; for a slowing body a is negative.
- Reading the slope of an x-t graph as acceleration and the slope of a v-t graph as velocity; slope of x-t is velocity, slope of v-t is acceleration.
- Calculating the distance from a v-t graph by adding signed areas or by taking only one segment; add the areas of all segments (triangles and rectangles) and take area below the axis as positive for distance.
- Thinking that acceleration is zero at the top of a vertical throw; only the velocity is zero, acceleration is still g.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 5 m/s squared	20/4.
2	**(d)** pi r	Half the circumference.
3	**(a)** 10 m/s	v squared = 2as = 100.
4	**(d)** 40 m	Triangle: 0.5 x 4 x 20.
5	**(d)** 10 s	15 m/s; 150/15.
6	**(b)** 45 m	0.5 x 10 x 9.
7	**(d)** (2n - 1)	Ratio 1 : 3 : 5.
8	**(d)** 20 m/s	200/10.
9	**(b)** 30 m/s	gt.
10	**(a)** 1 : 3 : 5	Odd numbers.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Slope = (60 - 20)/(6 - 2) (1); = 10 m/s (1).	1+1
12	10 m/s (1); 36 km/h (1).	1+1
13	v = 20 m/s (1); s = 75 m (1).	1+1
14	Zero (1); Net displacement is zero (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	u = 10, v = 20 m/s (1); a = 1 m/s squared (1); s = 150 m (1).	1+1+1
16	Uniform velocity 5 m/s (1); Uniform acceleration while rising (1); Then uniform velocity 10 m/s (1).	1+1+1
17	80/2 = 40 km/h (1); Distances in equal times unequal (1); Non-uniform (1).	1+1+1
	OR: s3 - s2 = 0.5 x 4 x (9 - 4) (1); = 10 m (1); Using the nth second formula (1).	1+1+1
18	Distance = 200 + 300 + 400 = 900 m (1); Time = 30 s (1); Average speed 30 m/s (1).	1+1+1
	OR: v = 3 m/s (1); s = 9 m (1); Formulas used (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Acceleration first part 2 m/s squared (1); Distance first part 16 m (1); Second part 32 m (1); Third part 8 m (retardation 4) (1); Total 56 m; average velocity 5.6 m/s (1).	1+1+1+1+1
	OR: s proportional to t squared (1); Starts from rest (1); s = 5t squared, so a = 10 m/s squared (1); v = at = 40 m/s at 4 s (1); Free fall (1).	1+1+1+1+1
20	Relative speed = 100 km/h (1); Time = 120/100 = 1.2 h (1); From A: 72 km (1); From B: 48 km (1); Total 120 km (1).	1+1+1+1+1
	OR: Relative speed = 150 km/h (1); = 41.7 m/s (1); Time = 300/41.7 (1); = 7.2 s (1); Opposite directions: speeds add (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** uniform acceleration	Speed increases.
2	**(c)** 25 m/s	90 x 5/18.
3	**(b)** 1 : 4 : 9	Proportional to t squared.
4	**(a)** 0.5 m/s squared	5/10.
5	**(b)** 10 m/s	100/10.
6	**(a)** 3 s	u/g.
7	**(d)** 5 m/s squared	100/20.
8	**(b)** zero	Net displacement zero.
9	**(a)** 45 m	u squared/2g.
10	**(c)** its velocity changes by equal amounts in equal times	Definition.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rising line (1); Horizontal line then falling line to the origin (1).	1+1
12	Distance 14 m (1); Displacement 10 m (1).	1+1
13	a = -3 m/s squared (1); s = 37.5 m (1).	1+1
14	Relative speed 5 m/s (1); Distance = 50 m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	u = 25 m/s (1); t = 20 s (1); s = 250 m (1).	1+1+1
16	h = 0.5 x 10 x 6.25 = 31.25 m (1); v = 25 m/s (1); Starts from rest (1).	1+1+1
17	a = 2 m/s squared (1); s = 15 x 10 = 150 m (1); Average velocity 15 m/s (1).	1+1+1
	OR: Distance 110 m (1); Displacement 30 m north (1); Average speed 2.2 m/s (1).	1+1+1
18	Speed = 4 m/s (1); Diameter = 20/3.14 = 6.37 m (1); Average velocity = 6.37/2.5 = 2.55 m/s (1).	1+1+1
	OR: Velocity direction changes (1); Change in velocity points to the centre (1); Hence centripetal acceleration (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Relative speed constant at 40 m/s (1); Time = 80/40 = 2 s (1); First stone falls 20 m (1); They meet 60 m above the ground (1); Check second: 80 - 20 = 60 (1).	1+1+1+1+1
	OR: A: 8 m/s = 28.8 km/h (1); B: 8.33 m/s = 30 km/h (1); When B finishes, A has run 96 m (1); B wins by 4 m (1); Calculation of A's distance 8 x 12 (1).	1+1+1+1+1
20	s_n = u + a(2n - 1)/2 (1); Substitute (1); = 2 + 4 x 9/2 (1); = 20 m (1); Check with s5 - s4 (1).	1+1+1+1+1
	OR: Area under graph = rectangle + triangle (1); ut + (1/2)at squared (1); = 20 + 25 (1); = 45 m (1); Verify using average velocity 9 x 5 = 45 (1).	1+1+1+1+1

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